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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSD415 MAIN PROJECT – FINAL REVIEW

MINDPHASE-M: A MULTILINGUAL CHATBOT FINE-TUNED WITH LORA ADAPTERS

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COMPUTER SCIENCE

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Introduction and Motivation

- 1 in 5 young adults experience mental health issues; most lack access to affordable, stigma-free support
- Existing chatbots are stateless — they forget previous conversations, breaking therapeutic continuity
- Cultural gap: Most tools are English-only
- Women's mental health is often linked to menstrual cycles — yet no chatbot integrates period/mood tracking
- Motivation: Build a low-resource, memory-aware, culturally sensitive companion that bridges emotional + physical health

Problem definition and Objectives

Problem Definition

Current mental health chatbots have several limitations, including a lack of persistent conversational memory that prevents continuity in user interactions, limited multilingual support that restricts accessibility for non-English speakers, no integration of physical health indicators such as sleep or activity data for holistic assessment, and a tendency to provide generic responses with weak personalization that reduces emotional engagement and effectiveness.

Research Objectives

1. Design **dual-memory architecture** (Slot + Episodic)
2. Enable **multilingual interaction** (EN / HI / ML)
3. Integrate **period tracking & mood analytics**
4. Implement **safe, privacy-aware system**

LITERATURE REVIEW

Sl. No.	Title of the paper	Journal Name& publication year	Algorithm / ToolsUsed	Advantages	Disadvantages
1	An AI-driven web application for mental health self-assessment with secure user management and intelligent chatbot support	IJERT, 2025	AI-driven web app, chatbot integration, secure user management	Provides self-assessment, integrates chatbot for support, secure user data handling	Limited to web app, requires internet, initial dataset bias
2	AI chatbots for mental health: Values and harms from lived experiences of depression	arXiv, 2025	Qualitative study, AI chatbots, lived user experiences	Highlights both values (support, availability) and harms (miscommunication, dependency)	Lack of clinical validation, risk of misinterpretation
3	Combining Artificial Users and Psychotherapist Assessment to Evaluate LLM-based Mental Health Chatbots	arXiv, 2025	LLMs, artificial users, therapist evaluation	Novel evaluation method, combines automated and expert assessments	Artificial users may not fully reflect real behavior, therapist availability limitsy
4	Evaluating an LLM-Powered Chatbot for Cognitive Restructuring	arXiv, 2025	LLM-based chatbot, cognitive restructuring framework	Provides professional insights, evaluates restructuring effectiveness	Limited testing scope, depends on professional feedback only
5	Chatbots and conversational agents in mental health: A review of the psychiatric landscape	Canadian Journal of Psychiatry, 2019	Literature review of multiple chatbots, psychiatric evaluation	Comprehensive overview of mental health chatbots, early trends identified	Outdated compared to recent LLM-based systems
6	IOverview of Chatbot-Based Mobile Mental Health Appsg	JMIR Mhealth Uhealth, 2023	Mobile apps, chatbot analysis, user reviews	Analyzes effectiveness from real users, practical insights	Limited to app store descriptions and reviews
7	FABRICATOR: Generative agents for synthetic emotions with emotional blending and contextual memory	arXiv, 2024	Generative AI, synthetic emotions, contextual memory	Advances emotional simulation in chatbots	Focused on synthetic emotions, not validated in clinical mental health
8	Using AI chatbots to provide self-help depression interventions for university students: A randomized trial	Internet Interventions, 2022	AI chatbot, randomized control trial (RCT)	Clinically validated, effective in reducing depression symptoms	Limited to student population, small sample size
9	LoRA: Low-rank adaptation of large language models	arXiv, 2021	ML models (Faster-RCNN, YOLO-F, DINO)	Precise teat shape/condition detection; runs on standard devices	Small dataset; only teat health; needs scaling
10	DyLoRA: Parameter-efficient tuning of pretrained models	arXiv, 2023	LoRa wireless sensors	Long-range, maintenance free, remote monitoring	Limited AI analytics; possible signal issues
11	GoRA: Gradient-driven adaptive low-rank adaptation	arXiv, 2025	InceptionTime deep learning + IMU sensors	Accurate lameness detection; fast online use	Focused on one axis; needs broader validation
12	Zero-shot and few-shot learning with generative AI	IJRAH, 2025	RT-DETR Transformer on CCTV video	Real-time detection; no extra hardware; high accuracy	Needs powerful hardware; limited starting dataset
13	Artificial Intelligence-Enabled Chatbots in Mental Health: A Systematic Review	Computers, Materials & Continua, 2022	Systematic Review of AI-enabled chatbots (Woebot, Wysa, Tess, etc.)	Broad survey of chatbot technologies, identifies strengths and limitations	Lacks experimental validation, no new model proposed
14	Chatbots and Stigma in Schizophrenia: The Need for Transparency	Schizophrenia Bulletin, 2024	AI chatbot for students, ML + NLP-based conversational agent	Highlights ethical issues, stigma reduction potential	Lack of technical model detail, emphasizes ethics only
15	Artificial Intelligence-Based Chatbot for Student Mental Health Support	Open Access Library Journal, 2024	AI chatbot for students, ML + NLP-based conversational agent	Provides targeted support for students, accessible mental health help	Limited testing population (students), not clinicily validated
16	Artificial Intelligence in Positive Mental Health: A Narrative Review	Indian Journal of Psychiatry, 2024	Narrative review of AI in positive psychology (chatbots, LLMs mentioned generally)	Summarizes AI contributions to well-being, positive mental health	Broad overview, no specific chatbot model implementation

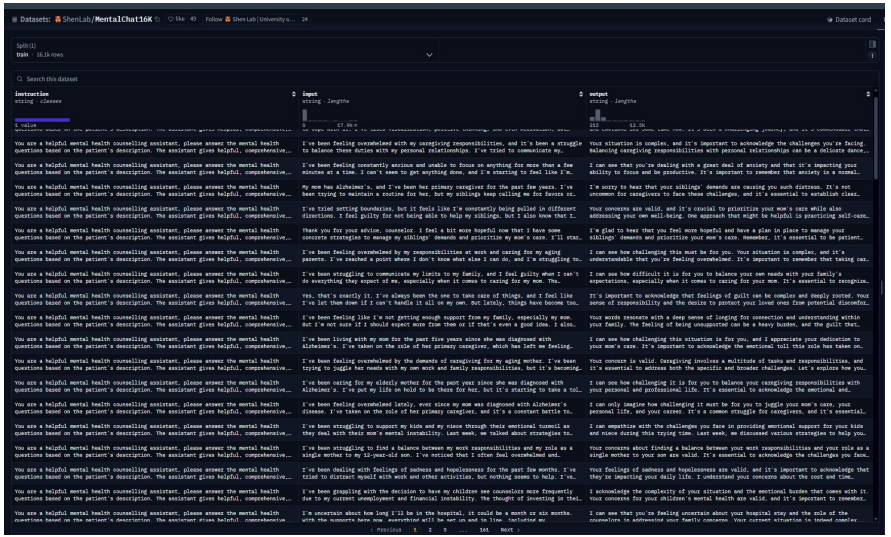
Dataset & Data Pipeline

Datasets Used

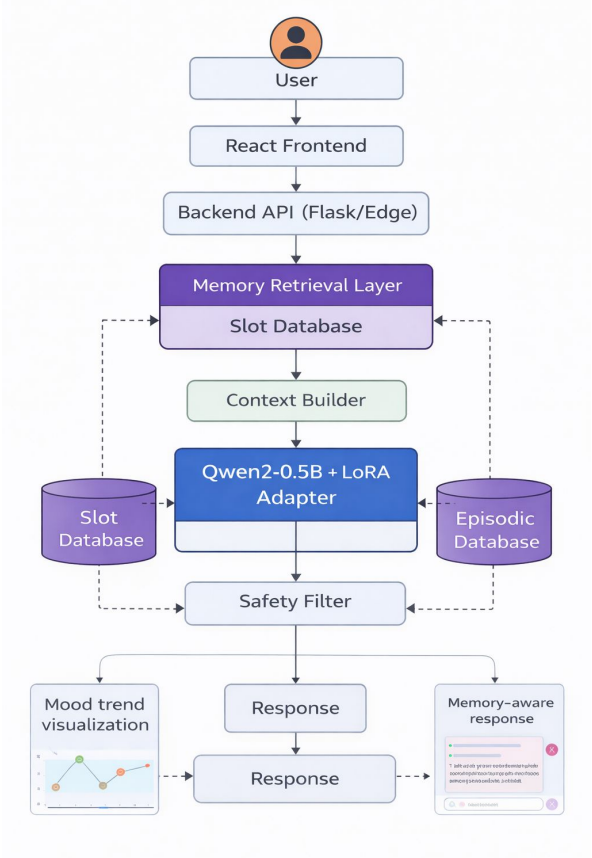
- Dataset Name: [ShenLab/MentalChat16K]
  - Dataset Size: ~16,000 conversation samples
  - Top-Level Attributes: 2 (id, conversations)
  - Nested Attributes: 2 (from, value)
- **EPITOME** — Motivational Interviewing patterns
- **Custom Slot Dataset** — User facts (name, triggers, preferences)
- **User Health Logs** — Mood & period entries (real-time data)

Data Pipeline

1. Data Collection & Cleaning
2. Dialogue Formatting (User–Assistant pairs)
3. Memory Extraction (slot + event summary)
4. Storage in PostgreSQL (RLS enabled)
5. Dynamic Retrieval during response generation



System Architecture



Memory Architecture

1. Memory Module

This module allows the chatbot to remember and personalize conversations.

- **Slot Memory:** Stores key user details like name, preferences, or recurring concerns.
- **Episodic Memory:** Stores summarized past interactions to maintain conversation continuity.

2. Health & Context Integration Module

This module improves emotional awareness and response relevance.

- Tracks user mood patterns and recurring stress indicators.
- Injects relevant past context into prompts before generating responses.

3. Response Control Module

This module ensures safe, concise, and meaningful replies.

- Manages prompt length to avoid overflow.
- Detects sensitive keywords for safer responses.
- Updates memory after each conversation.

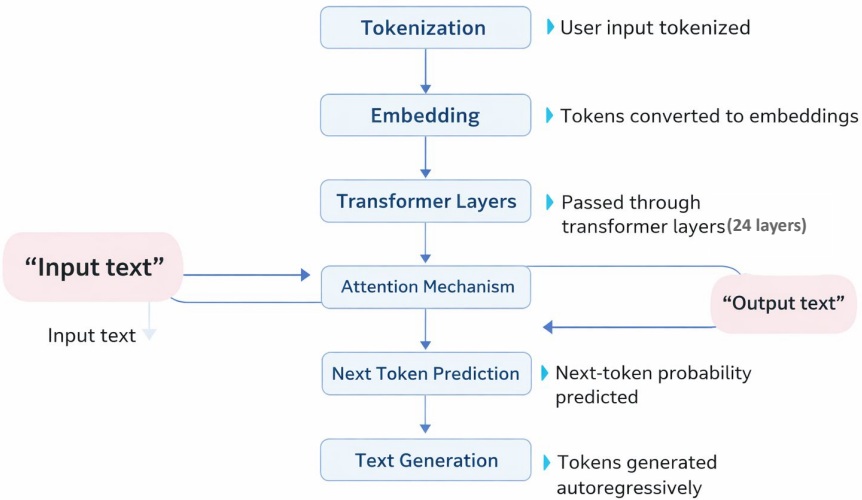
Model Architecture

Base Model

- Qwen2-0.5B Transformer
- Decoder-only architecture
- Multi-head self-attention
- RMSNorm + SwiGLU activation
- Byte-level tokenizer

How Response is Generated

1. User input tokenized
2. Tokens converted to embeddings
3. Passed through transformer layers
4. Attention computes contextual relevance
5. Next-token probability predicted
6. Tokens generated autoregressively



Training Strategy & Model Adaptation

Fine-Tuning Method

- PEFT (Parameter Efficient Fine Tuning)
- LoRA adapters added to:
  - Q_proj : Represents what the model is looking for.
    i. Equation: Q= XWq
  - K_proj : Represents what information each word contains.
    i. Equation: K= XWk
  - V_proj : Represents the actual meaning/content of each word.
    i. Equation: V= XWv
  - O_proj: Represents the final combined information after attention.
    i. Equation: Attention(Q,K,V)
- **Attention Equation :**

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

- **Q** = Query matrix
- **K** = Key matrix
- **V** = Value matrix
- **d** = Dimension of key vectors

Why LoRA?

- Only ~1–2% parameters trained
- Lower GPU usage
- Faster convergence
- Base model remains frozen

Training Setup

- 3 epochs
- Max length 256
- Learning rate 2e-5
- Causal LM objective

Model & Prompt Engineering Strategy

Model Architecture

- LLM:Qwen2-0.5B
- Cloud-based inference (low local computation)
- Streaming responses for real-time interaction

Prompt Engineering Strategy

- User slot memory (name, preferences)
- Relevant episodic summaries
- Current mood & cycle data
- Safety & empathy instructions

Response Control

- Context-length management
- Crisis keyword filtering
- Memory update after each interaction

Technology stack

Frontend

- React 18 + TypeScript
- Vite + Tailwind CSS
- shadcn/ui + Radix UI
- Recharts (mood visualization)

Backend & Database

- Cloud (Deno Edge Functions)
- PostgreSQL with Row-Level Security (RLS)
- Secure Email Authentication

AI Model Layer

- LoRA-based fine-tuning for empathy
- Parameter-efficient adaptation (PEFT)
- Lightweight, low-resource deployment

Experimental Results

System Performance Metrics

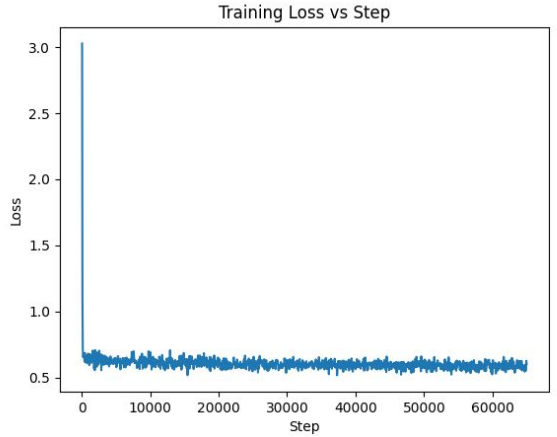
- Avg Response Time: **1.2 – 1.8 sec**
- Avg Response Length: **65–110 tokens**
- Memory Retrieval Latency: **< 120 ms**
- Context Injection Size: **~15–25% of total prompt**

Functional Validation

- Slot memory persists across sessions
- Episodic recall improves contextual relevance
- Mood-aware replies reflect recent journal entries
- Controlled response verbosity achieved

Observed Improvements

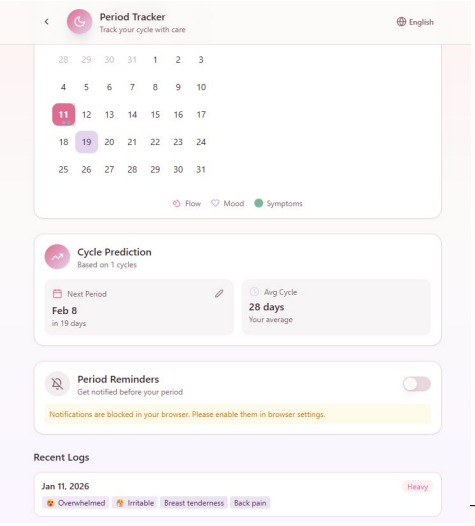
- Reduced repetitive responses
- Increased personalization quality
- More coherent multi-turn conversations



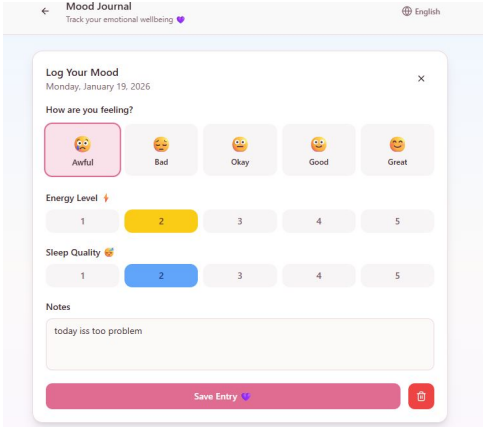
The graph shows that the training loss decreases quickly and then stabilizes, indicating the model has learned effectively and converged to consistent performance."

Comparative Evaluation

Feature	Mindphase-M	Woebot	Wysa
Long-term Memory	✓ Slot +Episodic	✗	✗
Multilingual Support	EN/HI/ML	English only	Limited
Period Tracking	✓ Integrated	✗	✗
Mood Analytics	✓ Trend charts	Basic	✓
Guided Exercises	1)Breathing Exercise 2)Ground Exercise 3) Muscle Exersise	CBT Modules	Meditation
Conversation History	✓	✗	✗
Data Privacy Control	Cloud - Based	Cloud - Based	Cloud-Based
Model Type	Fine-tuned Qwen 2-0.5B	Rule-Based	Hybrid AI
Avg Response Timesec	1.8 – 2 sec	~2 – 3 sec	~2 – 3 sec
Avg Response Length	65 – 110 tokens	Short scripted	Moderat

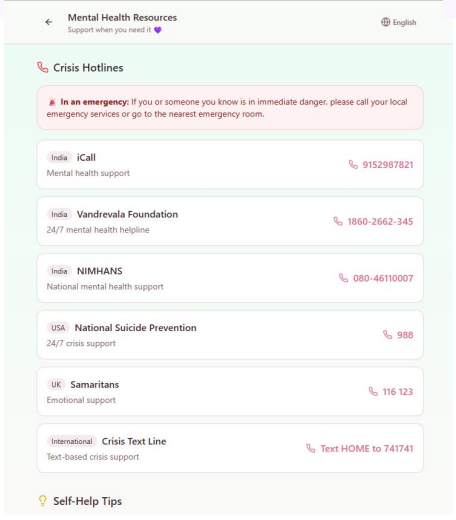


⇦ **Period Tracker**

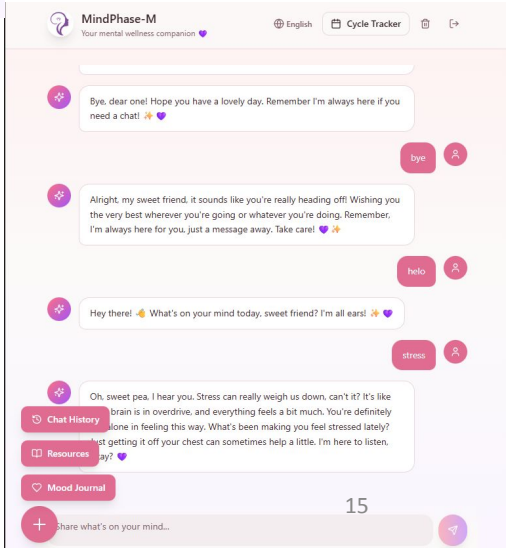


⇦ **Mood Journal**

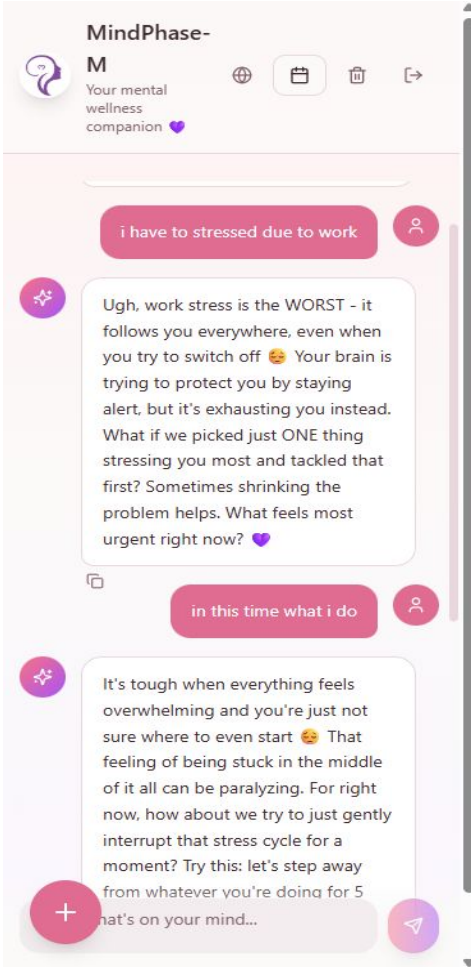
Crisis Hotlines ⇨



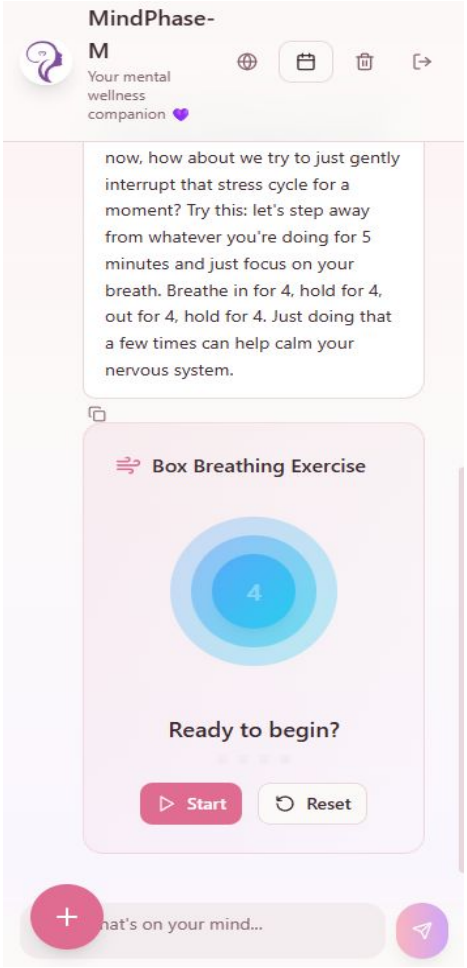
Chat Interface ⇨



Outcomes

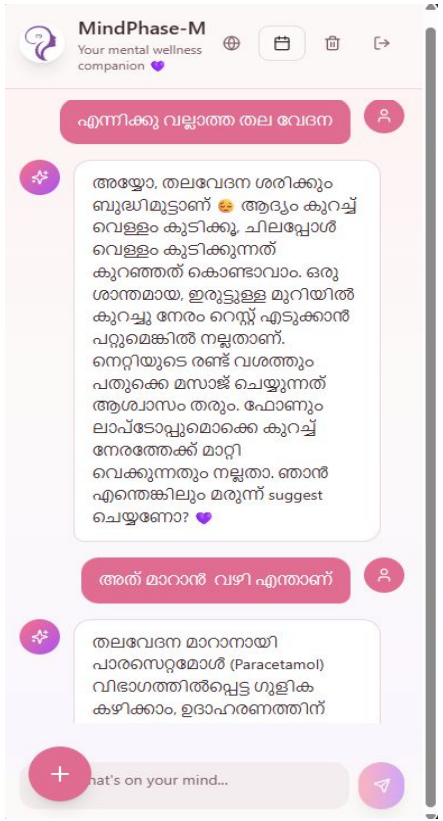


⇦ **English** ⇨



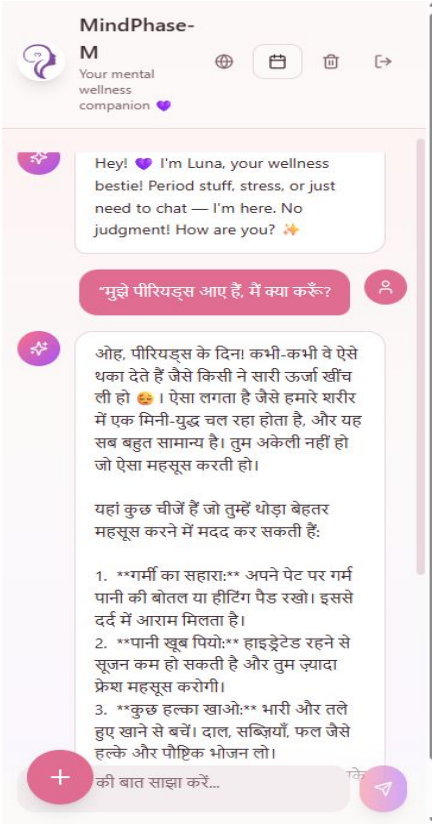
Outcomes

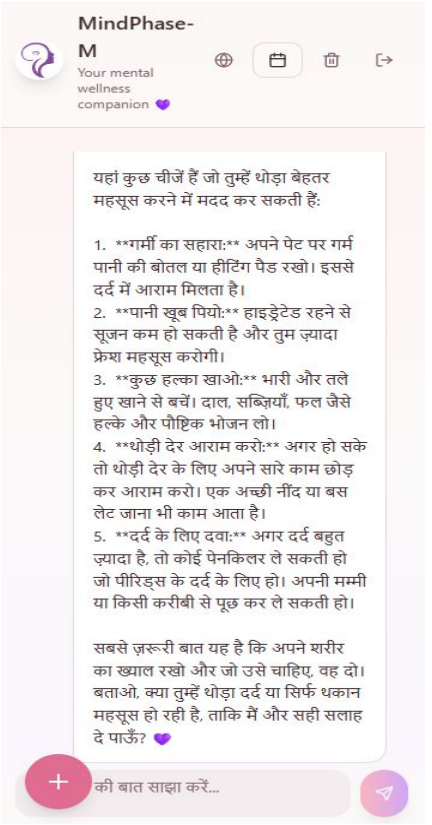
Malayalam





Hindi





Conclusion & Future work

Conclusion

- Developed a **memory-augmented mental health chatbot**
- Integrated **emotional + physical health tracking**
- Enabled **multilingual conversational support**
- Achieved controlled, personalized responses using fine-tuned Qwen2-0.5B
- Designed a secure, scalable cloud architecture

Future Work

- Advanced emotion classification adapter
- Proactive mood trend analysis
- Therapist monitoring dashboard (with consent)
- Voice interaction (speech-to-text & TTS)
- Lightweight on-device deployment

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THANK YOU